

Individual or Teamwork of 2		
Student(s) identification		
Al Labade	Ayman	

6 Points	Title of Incident:	Snowflake Cloud Data Platform Breach	
	Discovery date:	May 23, 2024	
	Start date of Incident:	April 14, 2024	
	End date of incident:	June 30, 2024 (ongoing monitoring)	
	Organization affected:	Snowflake Inc. (cloud data platform provider)	
		Number of employees	Approximately 7000 employees
		Revenues	\$2.67 billion (fiscal year 2024)
		Country	United States
		Line of business	Cloud-Based data warehousing and analytics platform
4	Cause of the incident:	Voluntary (deliberate cyberattack using stolen credentials)	
	Aspect affected:	Confidentiality, Integrity, and Availability	
10	Incident description		
	The UNC5537/ShinyHunters threat group conducted a large-scale credential-based cyberattack in April 2024 targeting Snowflake’s cloud platform customers. Using stolen credentials from previous malware infections dating to 2020, attackers accessed over 165 organizations’ data through accounts lacking multi-factor authentication. Major victims included AT&T (242M customers), Ticketmaster (560M customers), Santander Bank (30M customers), Advance Auto Parts (2.3M customers), and Neiman Marcus (65K customers). The attackers deployed a custom tool called “RapeFlake/FROSTBITE” to systematically breach accounts, ultimately compromising hundreds of millions of individuals personally identifiable information, financial records, medical data, and over 50 billion call records across multiple industries.		
10	Direct impacts		
	The Snowflake breach caused widespread operational chaos across more than 165 organizations in sectors like telecommunications, entertainment, finance, automotive, and healthcare. Many companies only learned of the incident after being notified by external parties, prompting immediate activation of emergency response protocols. The breach interrupted essential workflows, such as customer management, financial operations, billing, and data analytics. Due to reliance on Snowflake’s platform, these organizations had to quickly meet legal requirements under regulations like HIPAA, GDPR, CCPA, leading to a large-scale legal and notification efforts. The incident shook industry trust in cloud data platform security and let to service outages and major disruptions as affected businesses investigated and contained the breach. Companies also faced significant administrative burdens communicating with stakeholders and regulators, and many rapidly increased their cybersecurity defenses, introducing measures like mandatory multi-factor authentication and enhanced monitoring.		
5	Direct cost		

	Across Snowflake and its 165+ affected customers, the 2024 breach drove over \$1 billion in direct costs, including at least \$500K in forensics and consulting fees (Mandiant and CrowdStrike), seven-figure legal and compliance expenses for GDPR/HIPAA/CCPA obligations, \$250-\$750M for large-scale customer notifications and credit monitoring, millions on cybersecurity upgrades, and substantially higher cyber-insurance premiums.
10	Incident timeline
	• April 14, 2024: UNC5537 threat actors begin systematic exploitation of compromised Snowflake customer credentials. Initial successful intrusion into Advanced Auto Parts Snowflake environment, beginning 40-day period of unauthorized access. • April-May 2024: Attackers use “RapeFlake/FROSTBITE” tool to access multiple snowflake customer accounts lacking MFA protections. • May 23, 2024: Snowflake officially acknowledges potential unauthorized access to customer accounts and begins customer notification process. • June 2024: ShinyHunters lists 560 million Ticketmaster records and 30 million Santander customer files for sale on Breach forums cybercriminal marketplace. FBI seized Breach Forums on June 16, but attackers quickly established a replacement forum within 72 hours. • June 25, 2024: Neiman Marcus Group confirms Snowflake-related data breach affecting nearly 65,000 customers. • July 12, 2024: AT&T confirms that hackers accessed data from nearly all wireless customers through compromised Snowflake platform, affecting over 242 million customers and 128 million connected devices.
5	Perpetrator
	The Snowflake breach is attributed to the UNC5537 group (aka ShinyHunters) with very high confidence. This attribution rests on Mandiant’s detailed forensic analysis linking the incident to UNC5537’s unique tools, tactics, and infrastructure, as well as the groups deep familiarity with Snowflakes’ platform. The threat of actors’ public sale of stolen data further cements their responsibility. Lastly, ongoing FBI investigations into UNC5537’s activities and their demonstrated persistence, reconnaissance, and custom “RapeFlake/FROSTBITE” tooling reinforce the high certainty of this attribution.
5	Motivation
	The Snowflake breach was driven primarily by financial gain, executed through large-scale data theft and extortion, with very high confidence. UNC5537’s deliberate use of multi-pronged monetization tactics, such as demanding ransoms, selling bulk datasets on the dark net markets, and leveraging stolen records for identity fraud, demonstrating clear profit motivation.
10	Vulnerabilities being exploited and techniques used by the attackers
	The breach was enabled by basic gaps in credential management and cloud security practices rather than flaws in Snowflake itself. Attackers exploited over 80% of stolen credentials because companies hadn’t rotated passwords nor reinforced the need for multi-factor authentication. The attackers used a custom script to log in via SnowSight and SnowSQL and quietly siphoned off terabytes of data over weeks. On top of that, there were no IP allow-lists, spotty anomaly monitoring and no geo-restrictions, allowing them to move around freely without setting off any alarms.
10	Incident resolution

	Snowflake and its customers launched a coordinated, multi-step recovery effort: Snowflake immediately notified all affected clients, brought Mandiant and CrowdStrike for deep forensic digs, and executed mandatory multi-factor authentication plus advanced monitoring to spot abnormal access patterns. They also issued detailed compromise indicators and step-by-step guides for credential rotation and access reviews. Each of the 165+ customer organizations ran their own forensics, tightened security controls, and handled regulatory notifications under HIPAA, GDPR, CCPA, etc. While they locked out the intruders within weeks, full “closure” stretched over months because syncing efforts across so many companies and agencies took time. Although systems are back online, the damage cannot be undone as data is still floating around on the dark net markets and lingering identity-theft risks for millions.
5	Communications to external parties
	Snowflake’s communications were scattered in all directions. While the company quickly sent detailed technical alerts to its 165+ customers, it initially downplayed the breach as impacting only a limited number of accounts. That understatement hurt its credibility once the true scope emerged. Snowflake did have a communication framework, but reliance on each customer to handle public disclosures led to inconsistent messaging and confusion. The staggered, uneven disclosure approach fueled negative publicity and made it harder for stakeholders to grasp the incident’s full impact, ultimately undermining trust and complicating crisis management.
10	Long-term impacts
	The Snowflake breach is likely to produce lasting effects well beyond technical fixes. Reputational damage may hamper customer acquisition and retention as organizations tighten third-party risk assessments. Regulators will almost certainly impose stricter cloud-security mandates and new compliance frameworks around shared responsibilities and mandatory controls. Unfortunately, affected customers face prolonged legal exposure and hundreds of millions of individuals require ongoing identity-theft monitoring. Cyber-insurance premiums are set to rise sharply, and the incidents precedent may encourage similar attacks on other cloud platforms.
10	Lessons learned
	The Snowflake breach could have been largely prevented by enforcing non-negotiable security basics- mandatory multi-factor authentication, regular credential rotations, zero-trust principles, and strict IP/geolocation allow-listing with continuous monitoring and robust third-party risk assessments. While Snowflake’s platform security and swift incident response demonstrate partial due diligence, the failure to mandate MFA by default and the initial underestimation of the breach’s scope reveal significant gaps in security-by-design and customer visibility, indication that overall due diligence was incomplete.